AI-200 16-Week Study Plan — Pramod

Exam: Microsoft AI-200: Developing AI Cloud Solutions on Azure → Azure AI Cloud Developer Associate **Target:** March 2027 exam | **Start:** 15 Oct 2026 | **Plan ends:** 3 Feb 2027 | **Buffer:** 4 Feb – March for extra review + booking **Daily time:** 1 hour/day, flexible (early morning or late night) **Daily structure (60 min):** 25 min concept (simple language) + 20 min hands-on + 10 min quiz + 5 min recap **Weekly structure:** 6 study days + 1 review day (quiz + weak-area revisit) **Language:** Python | **Starting level:** Zero in Python + Azure basics | **No brain dumps** — Microsoft Learn + labs + trainer quizzes

How progress is tracked

- After each topic, you get a 5-question quiz. I log your score per domain.
- Each week ends with a review score: ✅ Mastered / 🟡 Shaky / 🔴 Revisit.
- Goal tracker: I update your AI-200 goal after each week. You can say "show my progress" anytime.
- Hands-on labs build toward 3 mini-projects + 1 capstone (see below).

Exam Domains (weights)

1. Develop AI Solutions Using Azure Data Services — 25-30%
2. Develop Containerized Solutions on Azure — 25-30%
3. Connect and Consume Azure Services — 20-25%
4. Secure, Monitor, and Troubleshoot Azure Solutions — 20-25%

PHASE 0 — Foundations (Weeks 1-2)

Week 1: Python from Zero

- Day 1: What is Python, why AI-200 uses it, install Python + VS Code, first program, print()
- Day 2: Variables, data types (str, int, float, bool), input(), type conversion
- Day 3: If / elif / else, comparison + logical operators, indentation
- Day 4: Loops — for, while, break/continue, range()
- Day 5: Lists, tuples, dicts, sets — the 4 collections you will use everywhere
- Day 6: Functions, parameters, return, scope, *args/**kwargs intro
- Day 7: Review + Quiz 1 (Python basics) + Fix mistakes
- Mini-lab: Build a small calculator + word counter CLI

- Sample quiz Qs: What does `len([1,2,3])` return? What is the difference between list and dict? When do you use `if` vs `elif`?

Week 2: Python Intermediate + Azure Basics + Tools

- Day 1: Strings, f-strings, string methods, file reading/writing
- Day 2: Error handling (try/except), modules, pip, virtual environments (venv)
- Day 3: OOP basics — classes, objects, **init**, methods (just enough for SDKs)
- Day 4: JSON, HTTP + REST in plain English, `requests` library, status codes
- Day 5: Git basics, GitHub, JSON/YAML, .env files, secrets — never hardcode keys
- Day 6: Azure basics — subscriptions, resource groups, portal tour, Azure CLI install, **create Azure Free Account** (only now, when we need it)
- Day 7: Review + Quiz 2 (Python + Azure basics) + Setup check
- Mini-lab: Python script that calls a public REST API and prints JSON nicely

PHASE 1 — Data for AI (Weeks 3-5) — 25-30%

Week 3: AI Concepts + Embeddings + Vector Search Intuition

- Day 1: What is AI vs ML vs GenAI, in simple words. Tokens explained.
- Day 2: Embeddings — words → numbers, why similar meanings = close numbers (visual diagram)
- Day 3: Distance metrics — cosine similarity vs Euclidean, with visual examples
- Day 4: Chunking — why we split documents, chunk size vs overlap trade-off
- Day 5: Vector search vs keyword search vs hybrid search vs semantic search
- Day 6: RAG in plain English — Retrieve → Augment → Generate, with flow diagram
- Day 7: Review + Quiz 3
- Lab: Use Python to compute cosine similarity between 3 sentences (no Azure needed)
- Sample quiz Qs: Why does RAG reduce hallucination? When is hybrid search better than pure vector search?

Week 4: Cosmos DB for NoSQL + Vector Search

- Day 1: NoSQL vs SQL, documents, containers, partition keys (why they matter)
- Day 2: Cosmos DB setup, create database/container in portal + Python SDK
- Day 3: Vector index policy, vector embeddings field, DiskANN vs quantized
- Day 4: Vector search queries in Cosmos DB, filtering + vector combined
- Day 5: Indexing policy, RU/s, throughput, cost control
- Day 6: Hands-on: Store 50 docs + embeddings, query top-5 similar
- Day 7: Review + Quiz 4

- Sample quiz Qs: What happens if you pick a bad partition key? What is RU/s?

Week 5: PostgreSQL + pgvector + Redis + RAG Project

- Day 1: Azure Database for PostgreSQL basics, pgvector extension
- Day 2: HNSW vs IVFFlat indexes — when to use which, with visuals
- Day 3: SQL + vector search combined — filter + similarity in one query
- Day 4: Azure Managed Redis — semantic caching, prompt caching, session storage
- Day 5-6: **Mini-Project 1 — RAG Q&A Bot:** Ingest docs → embed → store in pgvector/Cosmos → query → answer with grounding
- Day 7: Review + Quiz 5 + Project demo
- Sample quiz Qs: HNSW vs IVFFlat trade-offs? How does semantic caching save cost/latency?

PHASE 2 — Containers (Weeks 6-9) — 25-30%

Week 6: Docker Fundamentals

- Day 1: What is a container vs VM, images vs containers (diagram)
- Day 2: Dockerfile — FROM, COPY, RUN, CMD, layers
- Day 3: Build, run, tag, push, volumes, env vars, ports
- Day 4: Docker Compose — multi-container apps
- Day 5: Best practices — small images, .dockerignore, non-root user, secrets
- Day 6: Lab: Containerize your RAG API (FastAPI)
- Day 7: Review + Quiz 6
- Sample quiz Qs: What is the difference between CMD and ENTRYPOINT? Why keep images small?

Week 7: Azure Container Registry (ACR) + Azure Container Apps

- Day 1: ACR — private registry, repos, tagging, vulnerability scanning, Tasks (auto-build)
- Day 2: Container Apps — environments, revisions, ingress, scaling basics
- Day 3: Deploy your RAG API to Container Apps from ACR
- Day 4: Revisions + traffic splitting (blue/green), sidecar containers
- Day 5: KEDA autoscaling — scale on queue length, HTTP, CPU
- Day 6: Lab: Deploy v2, split 90/10 traffic, test rollback
- Day 7: Review + Quiz 7
- Sample quiz Qs: When to use revision vs new deployment? What does KEDA do?

Week 8: Kubernetes Basics + AKS

- Day 1: K8s in plain English — pods, deployments, services, namespaces (diagram)
- Day 2: YAML for K8s, kubectl basics
- Day 3: AKS — clusters, node pools, GPU node pools for AI inference
- Day 4: Deployments, rolling updates, Horizontal Pod Autoscaler (HPA)
- Day 5: Services, ingress, ConfigMaps + Secrets (no hardcoded secrets)
- Day 6: Lab: Deploy RAG API to AKS, test rolling update + HPA
- Day 7: Review + Quiz 8
- Sample quiz Qs: Pod vs Deployment vs Service? How does HPA decide to scale?

Week 9: Container Production + Project

- Day 1: Resource requests/limits, optimization, cost
- Day 2: Health probes (liveness/readiness), restart policies
- Day 3: Multi-arch images, image tagging strategy
- Day 4: Container Apps vs AKS — when to choose which (decision table)
- Day 5-6: **Mini-Project 2 — Containerized AI API:** ACR → Container Apps + AKS comparison deployment
- Day 7: Review + Quiz 9 + Phase 2 mock (15 Qs)

PHASE 3 — Events + Serverless (Weeks 10-12) — 20-25%

Week 10: Azure Functions

- Day 1: Serverless in plain English, Functions vs Container Apps
- Day 2: Triggers + bindings — HTTP, Timer, Blob, Cosmos DB, Service Bus, Event Grid
- Day 3: Build HTTP-triggered function in Python
- Day 4: Blob-triggered + Timer-triggered functions
- Day 5: Durable Functions intro (orchestration), cold start
- Day 6: Lab: Function that embeds new docs on blob upload
- Day 7: Review + Quiz 10
- Sample quiz Qs: Which trigger for "run every hour"? What is a binding?

Week 11: Event Grid + Event Hubs

- Day 1: Pub/Sub in plain English, events vs messages vs streams
- Day 2: Event Grid — routing, topics, subscriptions, serverless event processing
- Day 3: Event Hubs — high-throughput ingestion, partitions, consumer groups, telemetry
- Day 4: Event Grid vs Event Hubs vs Service Bus (decision table)
- Day 5: Lab: Blob → Event Grid → Function pipeline

- Day 6: Lab: Telemetry → Event Hubs → processing
- Day 7: Review + Quiz 11
- Sample quiz Qs: When to use Event Grid vs Event Hubs? What is a consumer group?

Week 12: Service Bus + Event-Driven Project

- Day 1: Queues vs Topics/Subscriptions, FIFO, ordering
- Day 2: Dead-letter queues, retry policies, duplicate detection
- Day 3: Transactions, sessions, message lock
- Day 4: Service Bus triggers in Functions
- Day 5-6: **Mini-Project 3 — Event-Driven AI Pipeline:** Upload → Event Grid → Function → Service Bus queue → worker → Cosmos DB
- Day 7: Review + Quiz 12 + Phase 3 mock

PHASE 4 — Security + Monitoring (Weeks 13-14) — 20-25%

Week 13: Identity + Secrets

- Day 1: Microsoft Entra ID — auth vs authz, OAuth, RBAC in plain English
- Day 2: Managed Identity — system-assigned vs user-assigned, secretless auth
- Day 3: Azure Key Vault — secrets, keys, certs, access policies
- Day 4: App Configuration, Key Vault references in Functions/Container Apps
- Day 5: Lab: Remove all hardcoded keys, switch to Managed Identity + Key Vault
- Day 6: Common exam traps — hardcoded secrets, over-permissive RBAC
- Day 7: Review + Quiz 13
- Sample quiz Qs: System vs user-assigned identity? Why is Managed Identity safer than connection strings?

Week 14: Monitoring + Troubleshooting with KQL

- Day 1: Azure Monitor + Application Insights — requests, dependencies, exceptions, availability
- Day 2: OpenTelemetry — distributed tracing, correlation IDs
- Day 3: KQL basics — where, project, summarize, sort, join (with examples)
- Day 4: KQL for troubleshooting — find slow requests, errors by operation
- Day 5: Logs, metrics, alerts, dashboards
- Day 6: Lab: Instrument your RAG API, write 5 KQL queries to debug a failure
- Day 7: Review + Quiz 14 + Phase 4 mock
- Sample quiz Qs: Write KQL to find top 5 slowest requests. What is a correlation ID for?

PHASE 5 — Mocks + Exam Readiness (Weeks 15-16)

Week 15: Timed Mocks + Case Studies

- Day 1: Full timed mock 1 (60 Qs, 120 min simulation, then review)
- Day 2: Weak-area deep-dive 1 (lowest domain)
- Day 3: Case-study drills — multi-service architecture scenarios
- Day 4: Weak-area deep-dive 2
- Day 5: Mock 2 (focus on weak domains)
- Day 6: KQL + SDK syntax rapid review, flashcards
- Day 7: Review + Quiz 15 (mixed)

Week 16: Final Review + Booking

- Day 1: Architecture decision drills — Container Apps vs AKS, Cosmos vs Postgres, Event Grid vs Hubs vs Bus
- Day 2: Security + monitoring rapid review
- Day 3: Mock 3 (final), exam-day strategy — time management, flagging, elimination
- Day 4: Book exam for March, check ID + Pearson VUE / OnVUE requirements
- Day 5: Light review only, no new material
- Day 6: Rest + confidence check
- Day 7: Ready — buffer week starts

Capstone (across Weeks 12-14, polished in Week 15)

Production RAG System: Containerized FastAPI RAG service on Container Apps/AKS + Cosmos DB/Postgres vector search + Redis caching + Event-driven ingestion (Functions + Event Grid + Service Bus) + Managed Identity + Key Vault + App Insights + KQL dashboards. This is your portfolio project + exam revision tool.

Resources (official-first)

- Microsoft Learn — Course AI-200T00-A: Develop AI cloud solutions on Azure
- Microsoft Learn docs for each service (Cosmos DB, Container Apps, AKS, Functions, Service Bus, Event Grid, Event Hubs, Key Vault, App Insights, KQL)
- Azure Free Account (from Week 2) + Azure CLI + Python SDK docs
- Docker + Kubernetes official tutorials for hands-on
- Practice: Trainer quizzes in chat + MeasureUp (Microsoft's official practice-test partner) for final mocks — not brain dumps

Rules for us

- If you miss a day, we shift — no guilt, just reschedule.
- Tell me "I did Day X" or "quiz me" and I'll log it.
- Every quiz: I explain *why* the right answer is right and why the others are wrong.
- Videos/visuals: I'll link Microsoft Learn videos + draw diagrams in chat when a concept needs it.